



Mini-Circuits

HIGH GAIN, HIGH IP3

Wideband Amplifier

ZX60-V63+

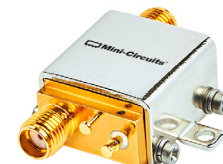
50Ω 0.05 to 6 GHz SMA Female

THE BIG DEAL

- High Gain
- Broadband High Dynamic Range
- Wideband, 0.05 to 6 GHz
- Protected by US Patent 6,790,049

APPLICATIONS

- Base Station Infrastructure
- Portable Wireless
- CATV & DBS
- MMDS & Wireless LAN
- LTE
- SATCOM
- Radar



Generic photo used for illustration purposes only

Model No.	ZX60-V63+
Case Style	GC957
Connectors	SMA female

+RoHS Compliant

The +Suffix identifies RoHS Compliance.
See our website for methodologies and qualifications

PRODUCT OVERVIEW

The ZX60-V63+ (RoHS compliant) uses Mini-Circuits' HBT technology to offer high gain over a broad frequency range and high IP3. Housed in a rugged, cost effective unibody chassis, this amplifier supports a wide variety of applications requiring moderate power output, low distortion, and 50Ω matched input/output ports.

KEY FEATURES

Feature	Advantages
High Gain 21.9 dB Typ. at 0.05 GHz 15.4 dB Typ. at 6 GHz	High gain reduces number of gain stages, at lower real estate, component count and cost. ±1.7 dB gain flatness from 50 MHz to 3 GHz.
Broadband, 0.05 to 6 GHz	Broadband covering primary wireless communications bands: Cellular, PCS, LTE, WiMAX, UHF, VHF, L band, Satcom, radar, etc.
High IP3 vs. DC Power Consumption +34.2 dBm Typical at 0.05 GHz +33.3 dBm Typical at 0.8 GHz	This model matches good IP3 performance relative to power consumption. The HBT structure provides good linearity over a broad frequency range as shown in the IP3 being typically 16 dB above the P1dB point to 0.8 GHz. This feature makes this amplifier ideal for use in: • Driver amplifiers for complex waveform upconverter paths • Drivers in linearized transmit systems
Very Small Size, 0.75x0.75"	The unique unibody construction enables the ZX60-V63+ to be used in compact designs.

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ELECTRICAL SPECIFICATIONS AT +25 °C AND +5.0 V, UNLESS NOTED OTHERWISE

Parameter	Condition (GHz)	Min.	Typ.	Max.	Units
Frequency Range		0.05		6	GHz
Gain	0.05	19.0	21.9		dB
	0.8		21.1		
	2.0		20.3		
	3.0		19.2		
	4.0		18.0		
	6.0		15.4		
Gain Flatness	0.05-3.0		±1.7		dB
	0.7-2.6		±1.3		
Input Return Loss	0.05	14.0	14.8		dB
	0.8		23.6		
	2.0		16.7		
	3.0		10.8		
	4.0		10.8		
	6.0		13.4		
Output Return Loss	0.05	12.0	15.7		dB
	0.8		15.5		
	2.0		13.6		
	3.0		15.9		
	4.0		24.1		
	6.0		11.8		
Output IP3	0.05		+34.2		dBm
	0.8		+33.3		
	2.0		+31.2		
	3.0		+28.8		
	4.0		+27.7		
	6.0		+23.9		
Output Power at 1 dB Compression	0.05	+17.0	+18.4		dBm
	0.8		+18.5		
	2.0		+17.8		
	3.0		+16.1		
	4.0		+15.0		
	6.0		+12.1		
Noise Figure	0.05		3.6		dB
	0.8		3.7		
	2.0		3.7		
	3.0		3.8		
	4.0		3.8		
	6.0		4.3		
Active Directivity (Isolation-Gain)	0.05-6		4.0		dB
DC Supply Voltage		+4.8	+5.0	+5.2	V
Supply Current		58	69	78	mA





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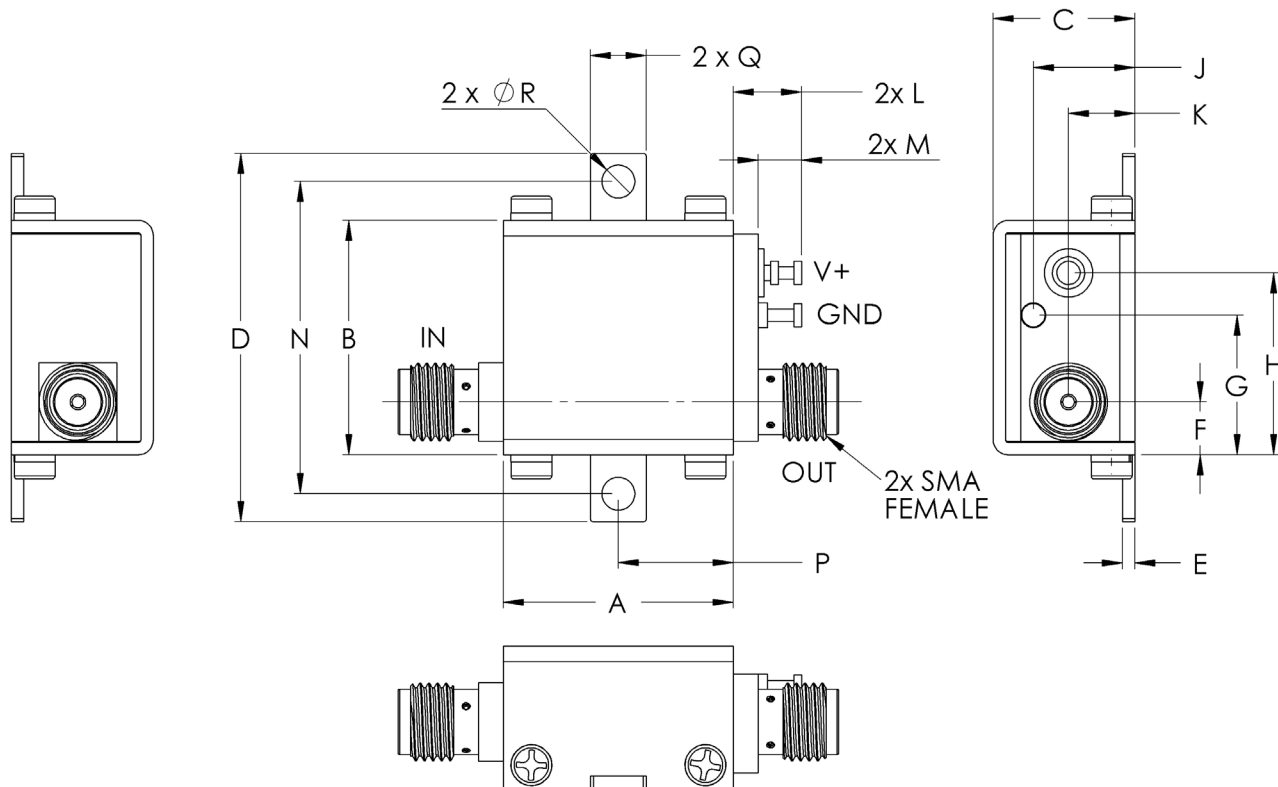
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ABSOLUTE MAXIMUM RATINGS

Parameter	Ratings
Operating Temperature	-40 °C to +85 °C Case
Storage Temperature	-55 °C to +100 °C
DC Voltage	+5.7 V
Input RF Power (No Damage)	+13 dBm
Power Consumption	0.5 W

Permanent damage may occur if any of these limits are exceeded.

OUTLINE DRAWING



NOTE: When soldering the DC connections, caution must be used to avoid overheating the DC terminal. See Application Note. [AN-40-010](#).

OUTLINE DIMENSIONS (Inches/mm)

A	B	C	D	E	F	G	H	J	K	L	M	N	P	Q	R	wt
.74	.75	.46	1.18	.04	.17	.45	.59	.33	.21	.22	.14	1.00	.37	.18	.106	grams
18.80	19.1	11.68	30.0	1.02	4.32	11.4	14.99	8.38	5.33	5.59	3.56	25.40	9.40	4.57	2.69	23.0





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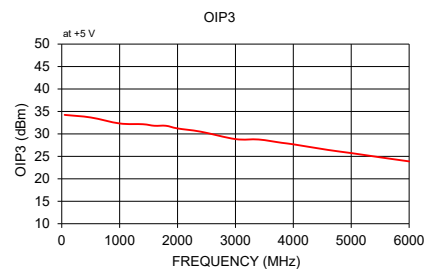
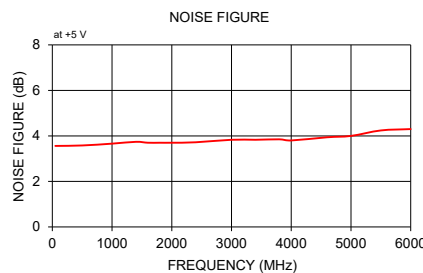
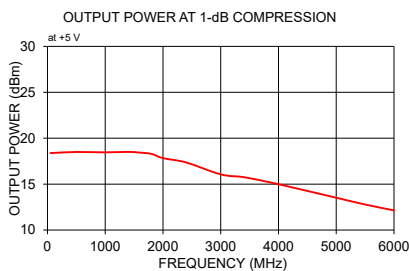
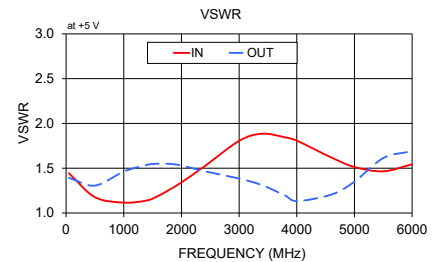
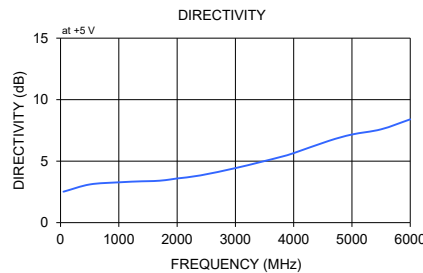
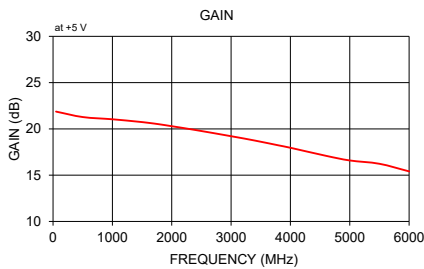
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TYPICAL PERFORMANCE DATA/CURVES

Frequency (MHz)	Gain (dB)	Directivity (dB)	VSWR (:1) +5 V		Power Out at 1 dB COMPR. (dBm)	Noise Figure (dB)	Output IP3 (dBm)
	+5 V	+5 V	IN	OUT	+5 V	+5 V	+5 V
50	21.87	2.52	1.44	1.39	18.4	3.6	34.2
500	21.28	3.10	1.18	1.31	18.5	3.6	33.7
1000	21.04	3.28	1.12	1.46	18.5	3.7	32.3
1400	20.80	3.36	1.14	1.54	18.5	3.7	32.2
1600	20.66	3.38	1.19	1.55	18.4	3.7	31.8
1800	20.49	3.46	1.26	1.55	18.3	3.7	31.8
2000	20.29	3.59	1.34	1.53	17.8	3.7	31.2
2400	19.88	3.84	1.52	1.47	17.4	3.7	30.5
3000	19.21	4.43	1.81	1.38	16.1	3.8	28.8
3400	18.74	4.89	1.89	1.31	15.8	3.8	28.8
3800	18.22	5.37	1.84	1.19	15.3	3.9	28.0
4000	17.96	5.65	1.81	1.13	15.0	3.8	27.7
4600	17.09	6.64	1.62	1.21	14.1	3.9	26.4
5000	16.59	7.17	1.51	1.35	13.5	4.0	25.7
5500	16.23	7.59	1.47	1.61	12.8	4.2	24.8
6000	15.41	8.40	1.54	1.69	12.1	4.3	23.9



NOTES

- Performance and quality attributes and conditions not expressly stated in this specification document are intended to be excluded and do not form a part of this specification document.
- Electrical specifications and performance data contained in this specification document are based on Mini-Circuit's applicable established test performance criteria and measurement instructions.
- The parts covered by this specification document are subject to Mini-Circuits standard limited warranty and terms and conditions (collectively, "Standard Terms"); Purchasers of this part are entitled to the rights and benefits contained therein. For a full statement of the standard. Terms and the exclusive rights and remedies thereunder, please visit Mini-Circuits' website at www.minicircuits.com/terms/viewterm.html

